

Problem 1. Find the equation of the tangent plane to the surface

$$x^2y + yz^2 = 5$$

at the point $(1, 1, 2)$.

Let $f(x, y, z) = x^2y + yz^2$, so the surface is $f(x, y, z) = 5$.¹ We compute the gradient field:

$$\nabla f(x, y, z) = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right\rangle = \langle 2xy, x^2 + z^2, 2yz \rangle.$$

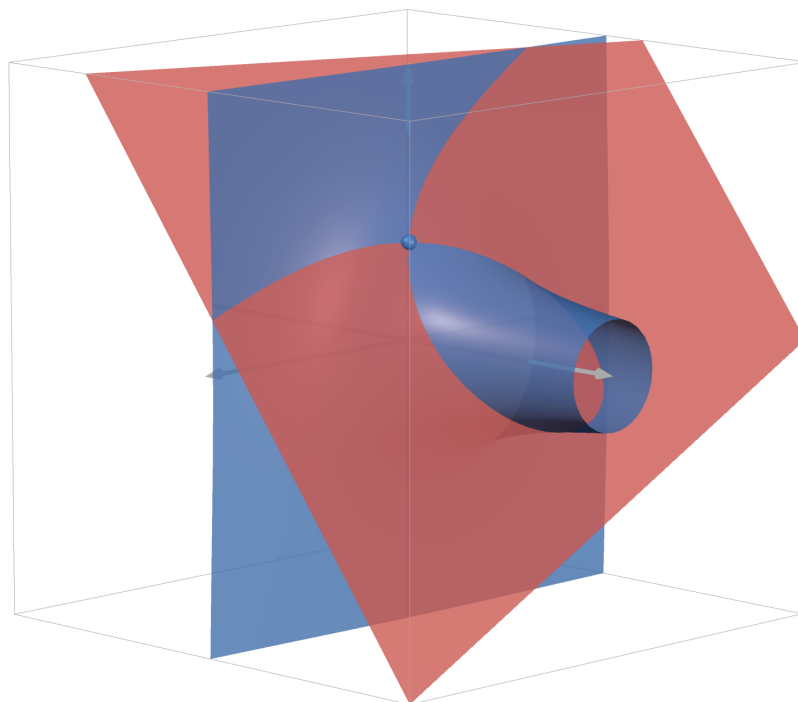
The gradient vector at the point $(1, 1, 2)$ is

$$\nabla f(1, 1, 2) = \langle 2(1)(1), (1)^2 + (2)^2, 2(1)(2) \rangle = \langle 2, 5, 4 \rangle.$$

The tangent plane contains the point $(1, 1, 2)$ and is perpendicular to the gradient vector $\langle 2, 5, 4 \rangle$, so the equation of the tangent plane is

$$\begin{aligned} \langle 2, 5, 4 \rangle \bullet \langle x - 1, y - 1, z - 2 \rangle &= 0 \\ 2(x - 1) + 5(y - 1) + 4(z - 2) &= 0 \\ 2x + 5y + 4z &= 15. \end{aligned}$$

Here is a picture:



¹It is also common to let $f(x, y, z) = x^2y + yz^2 - 5$ so the surface has the form $f(x, y, z) = 0$. It doesn't matter.

Problem 2. The scalar function $f(x, y) = x^4 - x^2 + y^2 + 2xy$ has three critical points:

$$(0, 0), \quad (1, -1), \quad (-1, 1).$$

Use the second derivative test to determine whether each of these is a local maximum, local minimum, or a saddle point.

To find the Hessian matrix we compute all first and second partial derivatives:

$$f_x = 4x^3 - 2x + 2y,$$

$$f_y = 2y + 2x,$$

$$f_{xx} = 12x^2 - 2,$$

$$f_{xy} = 2,$$

$$f_{yx} = 2,$$

$$f_{yy} = 2.$$

Hence the Hessian matrix is

$$Hf(x, y) = \begin{pmatrix} 12x^2 - 2 & 2 \\ 2 & 2 \end{pmatrix}$$

and its determinant is

$$\det Hf(x, y) = (12x^2 - 2)(2) - (2)(2) = 24x^2 - 8.$$

The critical point $(0, 0)$ has $\det Hf(0, 0) = -8 < 0$ so it is a saddle point. The critical points $(1, -1)$ and $(-1, 1)$ have $\det Hf(1, -1) = 16 > 0$ and $\det Hf(-1, 1) = 16 > 0$, so each of these is a local min or a local max. To see which, we can look at f_{yy} . Since $f_{yy}(x, y) = 2$ is always positive we conclude that each of $(1, -1)$ and $(-1, 1)$ is a local minimum.

Here is a picture:

